

$$\int \frac{dx}{\sqrt{x^2+c}} = \ln|x + \sqrt{x^2+c}| \quad c \neq 0$$

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{a}} \ln \left| ax + \frac{b}{2} + \sqrt{a} \cdot \sqrt{ax^2+bx+c} \right|$$

$$\int \frac{x^2+8}{\sqrt{x^2+16}} dx = \underbrace{(ax+b)}_{\text{dziwcy podkreślenie}} \sqrt{x^2+16} + \underbrace{A}_{\text{dziwcy podkreślenie}} \int \frac{1}{\sqrt{x^2+16}} dx \quad \parallel \cdot ()'$$

$$\frac{x^2+8}{\sqrt{x^2+16}} = a \sqrt{x^2+16} + (ax+b) \frac{2x}{2\sqrt{x^2+16}} + A \cdot \frac{1}{\sqrt{x^2+16}} \quad \parallel \cdot \sqrt{x^2+16}$$

$$x^2+8 = a(x^2+16) + (ax+b) \cdot x + A$$

$$x^2+8 = ax^2 + 16a + ax^2 + bx + A$$

$$x^2+8 = 2ax^2 + bx + 16a + 2b + A$$

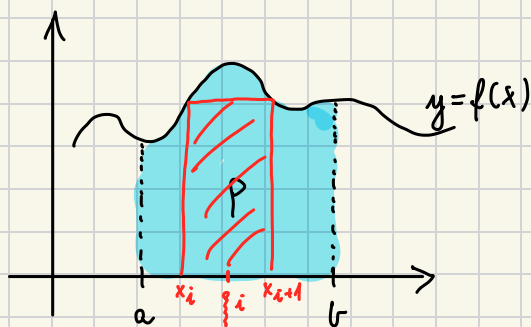
$$\begin{cases} 2a=1 \Rightarrow a=\frac{1}{2} \\ b=0 \\ 16a+2b+A=8 \end{cases}$$

$$8 + 2 \cdot 0 + A = 8$$

$$A=0$$

$$\begin{aligned}
 \int \sin^7 x \cdot \cos^4 x \, dx &= \int \sin^6 x \cdot \sin x \cdot \cos^4 x \, dx = \int (1 - \cos^2 x)^3 \cdot \sin x \cdot \cos^4 x \, dx = \\
 &= \left\{ \begin{array}{l} \cos x = t \\ -\sin x \, dx = dt \end{array} \right\} = \int (1 - t^2)^3 \cdot \sin x \cdot t^4 \cdot \frac{dt}{-\sin x} = - \int (1 - t^2)^3 \cdot t^4 \, dt = \\
 &= - \int (1 - 3t^2 + 3t^4 - t^6) t^4 \, dt = - \int (t^4 - 3t^6 + 3t^8 - t^{10}) \, dt = \\
 &= -\frac{1}{5} t^5 - \frac{3}{7} t^7 + \frac{3}{9} t^9 - \frac{1}{11} t^{11} + C = \\
 &= -\frac{1}{5} \cos^5 x - \frac{3}{7} \cos^7 x + \frac{1}{3} \cos^9 x - \frac{1}{11} \cos^{11} x + C
 \end{aligned}$$

CAŁKA OZNACZONA



$$\sum_{i=1}^n f(\xi_i) \cdot \Delta x$$

$$F'(x) = f(x)$$

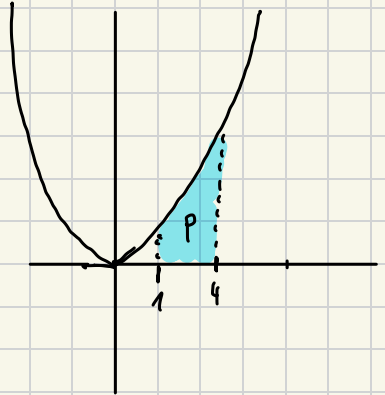
$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x = \int_a^b f(x) \, dx = F(b) - F(a)$$

Typy zadań dla całki oznaczonej

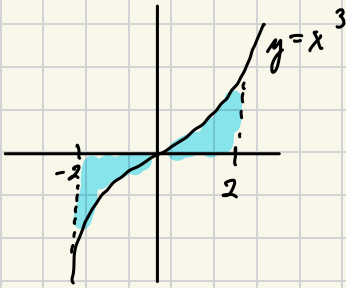
1) $\int_5^{10} x^4 dx = \frac{1}{5} x^5 \Big|_5^{10} = \frac{1}{5} \cdot (10)^5 - \frac{1}{5} (5)^5$

2) Oblicz pole obszaru ograniczonego osią OX , prostymi $x=1$, $x=4$ i parabolą $y=x^2$

$$P = \int_1^4 x^2 dx = \frac{1}{3} x^3 \Big|_1^4 = \frac{1}{3} (4)^3 - \frac{1}{3} (1)^3 = 21$$



3) Oblicz całkę $\int_{-2}^2 x^3 = \frac{1}{4} x^4 \Big|_{-2}^2 = 0$



4) Oblicz pole zamalowanego obszaru

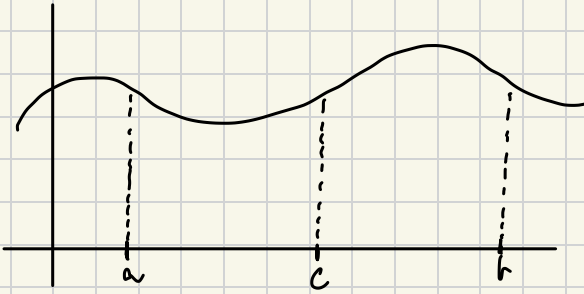
$$P = 2 \int_0^2 x^3 = 2 \cdot \frac{1}{4} x^4 \Big|_0^2 = \frac{1}{2} \cdot (2)^4 = 8$$

Catka oznaczona jest funkcją liniową

$$\int_a^b (f_1 \pm f_2) dx = \int_a^b f_1 dx \pm \int_a^b f_2 dx$$

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx$$

Catka oznaczona jest addytywna względem obszaru całkowania



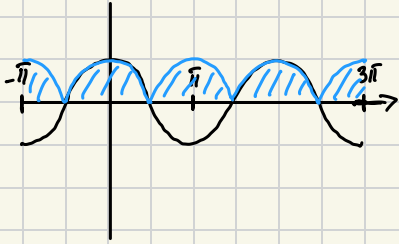
$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$5!! = 5 \cdot 3 \cdot 1$$

$$\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx = \begin{cases} \frac{(n-1)!!}{n!!} & ; n \text{ nieparzyste} \\ \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & ; n \text{ parzyste} \end{cases}$$

$$\int_{-\pi}^{3\pi} \cos^4 x \, dx = 8 \int_0^{\frac{\pi}{2}} \cos^4 x \, dx = \frac{3!!}{4!!} \cdot \frac{\pi}{2} = \frac{1 \cdot 3}{4 \cdot 2} \cdot \frac{\pi}{2} = \frac{3\pi}{16}$$

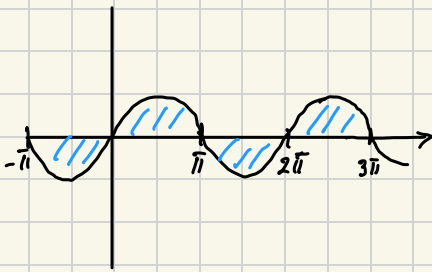


$$\int_0^3 x e^x \, dx = \left\{ \begin{array}{l} f = x \quad f' = 1 \\ g' = e^x \quad g = e^x \end{array} \right\} =$$

$$\int_{-\pi}^{3\pi} \sin^5 x \, dx = 0$$

$$= x \cdot e^x \Big|_0^3 - \int_0^3 e^x \, dx =$$

$$= x \cdot e^x \Big|_0^3 - e^x \Big|_0^3 = 2e^3 + 1$$



$$= \left\{ \begin{array}{l} t = x^2 \\ x = \sqrt{t} \\ \sqrt{t} = 2 \\ t = 4 \end{array} \right\} = t = 4$$

$$\int_0^2 x(x^2+3)^5 \, dx = \left\{ \begin{array}{l} x^2 + 3 = t \\ 2x \, dx = dt \\ dx = \frac{dt}{2x} \end{array} \right. \begin{array}{l} 0^2+3=3 \\ 2^2+3=7 \end{array} \right\} =$$

$$= \int_3^7 x \cdot t^5 \frac{dt}{2x} = \frac{1}{2} \int_3^7 t^5 \, dt = \frac{1}{2} (7^6 - 3^6)$$

$$a) \sum_{n=1}^{\infty} \frac{3\sqrt{n}+1}{4n^2+3n+2} \quad a_n \quad a_n \sim b_n = \frac{\sqrt{n}}{n^2} = \frac{n^{\frac{1}{2}}}{n^2} = \frac{1}{n^{\frac{3}{2}}} \quad \Delta = \frac{3}{2} > 1, \text{ więc } b_n \text{ zbieżny}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \left(\frac{3\sqrt{n}+1}{4n^2+3n+2} \cdot \frac{n^{\frac{3}{2}}}{1} \right) = \lim_{n \rightarrow \infty} \left(\frac{3n^2+n\sqrt{n}}{4n^2+3n+2} \right) = \frac{3}{4} \in (0; +\infty) \quad \text{szereg } \sqrt[n]{a_n} \text{ zbieżny}$$

to $\sqrt[n]{a_n}$ jest zbieżny różni (kryterium porównawcze graniczne)

$$b) \sum_{n=1}^{\infty} \left((-1)^{n+1} \cdot \left(\frac{2n+3}{5n-1} \right)^n \right) \quad a_n$$

Kryterium Conchezo

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| (-1)^{n+1} \cdot \left(\frac{2n+3}{5n-1} \right)^n \right|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2n+3}{5n-1} \right)^n} = \lim_{n \rightarrow \infty} \frac{2n+3}{5n-1} = \frac{2}{5} < 1, \text{ więc } \sum a_n < \infty$$

To możemy pominąć przez moduł